

作业1/4

T字形截面的悬臂梁，承受与杆轴平行的力F作用，如图所示。试求杆内的最大正应力。

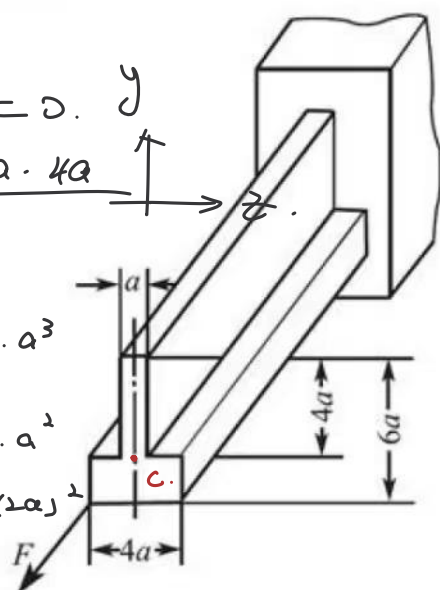
先计算截面形心。

由对称性得 $z_c = 0$ 。

$$y_c = \frac{a \cdot 2a \cdot 4a + 4a \cdot a \cdot 4a}{4a \cdot 2a + 4a \cdot a} = 2a$$

$$I_y = \frac{1}{12} \cdot 2a \cdot (4a)^3 + \frac{1}{12} \cdot 4a \cdot a^3 = 11a^4$$

$$I_z = \frac{1}{12} \cdot 4a \cdot (2a)^3 + 4a \cdot 2a \cdot a^2 + \frac{1}{12} \cdot a \cdot (4a)^3 + 4a \cdot a \cdot (2a)^2 = 32a^4$$



$$M_y = F \cdot 2a$$

$$M_z = F \cdot 2a$$

$$\begin{aligned} \sigma_{max} &= \frac{F}{A} + \frac{M_y \cdot z_A}{I_y} + \frac{M_z \cdot y_A}{I_z} \\ &= \frac{F}{10a^2} + \frac{4Fa^2}{11a^4} + \frac{4Fa^2}{32a^4} \\ &= 0.572 \frac{F}{a^2} \end{aligned}$$

作业2/4

试确定图所示截面的截面核心边界。

$$A = 5 \times 0.2^2$$

$$\begin{aligned} I_z = I_x &= \frac{0.2}{12} \times 0.6^3 + 2 \times \frac{0.2}{12} \times 0.2^3 \\ &= \frac{29}{1500} \text{ m}^4 \end{aligned}$$

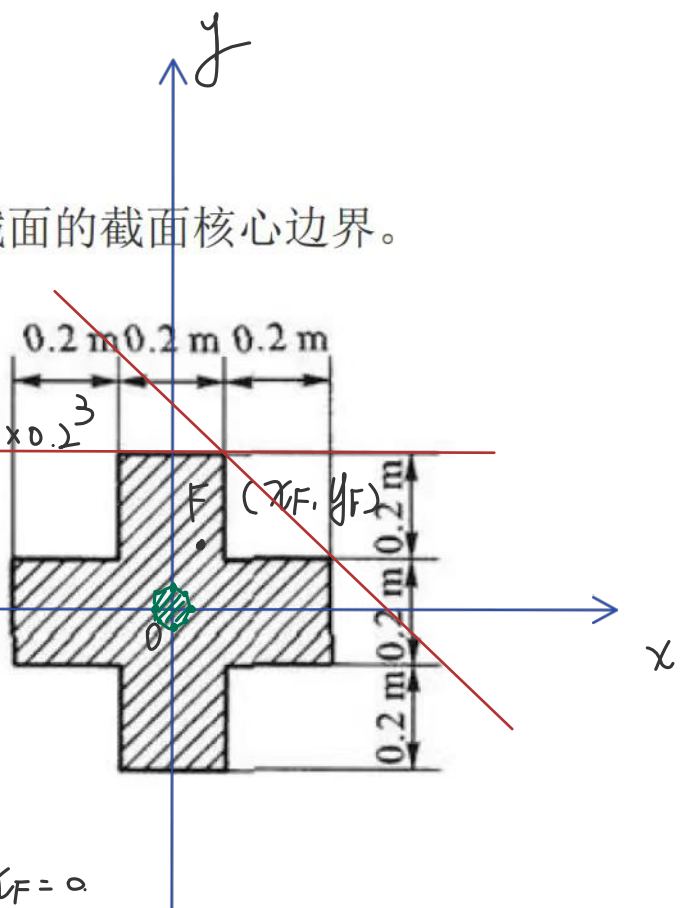
$$i_y^2 = i_x^2 = \frac{I}{A} = \frac{29}{1500} \text{ m}^2$$

当中性轴为 $y = 0.2$ 时。

$$\text{得到 } y_F = -\frac{29}{450} \text{ m}, \quad x_F = 0$$

$$\text{当中性轴为 } x + y = 0.4 \text{ 时 } y_F = -\frac{29}{600} \text{ m}, \quad x_F = -\frac{29}{600} \text{ m}.$$

图中八边形为核心区域



作业3/4

曲拐受力如图所示，其圆杆部分的直径 $d=50\text{mm}$ 。试画出表示A点处应力状态的单元体，并求其主应力及最大切应力。

$$F = 3.2\text{kN}$$

$$M_e = 3.2\text{kN} \times 140\text{mm} = 448\text{N}\cdot\text{m}$$

$$\sigma_A = \frac{M}{W} = \frac{32 \times 90}{\frac{\pi}{32} d^3}$$

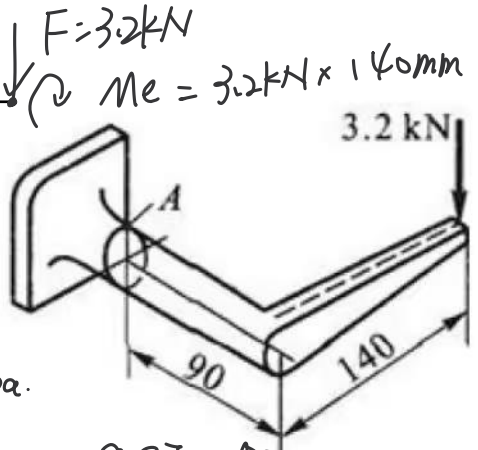
$$= 23.48\text{MPa}$$

$$\tau_A = \frac{T}{W_p} = \frac{16T}{\pi d^3} = 18.26\text{MPa}$$

$$\sigma_1 = \frac{\sigma_A}{2} - \sqrt{\frac{1}{4}\sigma_A^2 + \tau_A^2} = -9.97\text{MPa}$$

$$\sigma_2 = \frac{\sigma_A}{2} + \sqrt{\frac{1}{4}\sigma_A^2 + \tau_A^2} = 33.45\text{MPa}$$

$$\sigma_3 = 0$$

$$\tau_{\max} = \frac{|\sigma_1 - \sigma_2|}{2} = 21.71\text{MPa}$$


作业4/4

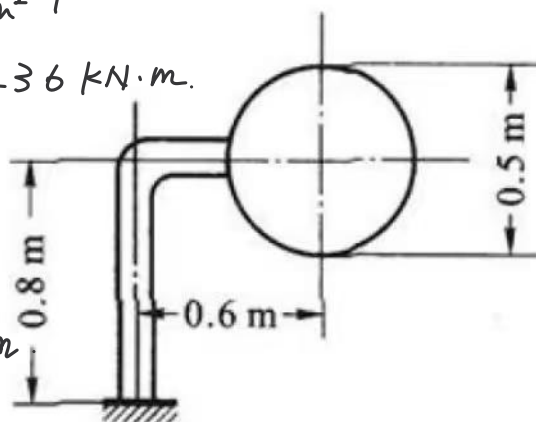
铁道路标圆信号板（如图所示），装在外直径 $D=60\text{mm}$ 的空心圆柱上，所受的最大风载 $p=2\text{kN/m}^2$ ，其材料的许用应力 $[\sigma]=60\text{MPa}$ 。试按第三强度理论选定空心柱的厚度。

$$F = p \cdot A = 2 \frac{\text{kN}}{\text{m}^2} \times \frac{\pi}{4} D^2 = 0.393\text{kN}$$

$$M_e = F \Delta x = 0.236\text{kN}\cdot\text{m}$$

在底部是危险截面

$$M_{\max} = F \times 0.8 = 0.314\text{kN}\cdot\text{m}$$



$$\sigma_{r3} = \frac{\sqrt{M_{\max}^2 + M_e^2}}{W} = \frac{\sqrt{M_{\max}^2 + M_e^2}}{\frac{1}{32} \pi (D^4 - d^4)} \leq [\sigma]$$

得 $d \leq 54.7\text{mm}$

$\therefore \delta = \frac{D-d}{2} = 2.65\text{mm}$

